

JAMES JAW

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EDUCATION

University of Wisconsin-Madison

M.S. in Computer Science

Sep 2023 – Dec 2025

GPA: 3.9

National Tsing Hua University

Post-Baccalaureate in Computer Science

Sep 2020 – Jan 2022

GPA: 3.9

SKILLS

Languages: Python, Golang, Java, C++, JavaScript, TypeScript, HTML/CSS

Frameworks: Django, Spring, React, Node.js, gRPC, REST APIs, PostgreSQL, MySQL, OpenAI API

Infrastructure: Docker, Kubernetes, GitHub Actions, AWS (EKS, EC2, S3, RDS, IAM)

EXPERIENCE

Open Source – Contributor

Feb 2026 – Present

KubeRay, Apache Airflow – *Golang, Python, TypeScript, Docker, Kubernetes*

- Diagnosed a KubeRay worker-startup bottleneck: right-sized the wait-gcs-ready init container's CPU/memory to match the Ray container, cutting up to **55s** of autoscaling latency per new worker
- Resolved a flaky e2e test caused by a concurrent-update race condition and modernized lint rules across the operator codebase; both merged upstream
- Contributed 5 merged PRs to Apache Airflow, including a React rendering fix for duplicate-key list reconciliation in the web UI and a declarative YAML migration of provider connection metadata across 3 provider packages

Digital Aid Seattle – Software Engineer

May 2026 – Present

Scheduling Platform – *TypeScript, Next.js, Supabase (Postgres/Auth/Edge Functions)*

- Built a full-stack scheduling tool that automates student grouping via timezone-aware availability matching, replacing manual spreadsheet workflows for multiple instructors
- Designed multi-tenant data isolation with Supabase Row Level Security, enabling instructors to independently manage class data on a shared platform

IBM – Software Engineer Intern

Jul 2025 – Aug 2025

Middleware Modernization – *C++, Shell, Docker, Kubernetes, Argo CD, Grafana*

- Built a C++ middleware integrating legacy CORBA APIs with modern deployment infrastructure, automating a previously manual validation workflow (**~60% faster**)
- Migrated legacy C++ services to Kubernetes, replacing manual deployments with Blue-Green GitOps releases
- Built team-friendly k6 gRPC load-testing tooling to simulate **1,000+** concurrent users and expose bottlenecks

PROJECTS

Video Chat Platform – *Golang, Fiber, TypeScript, React, WebRTC, Postgres*

2026

- Shipped BuckyChat, a campus-only video chat platform for verified UW–Madison students, as the sole engineer from zero to production
- Architected peer-to-peer WebRTC media over a lightweight Go signaling backend with Cloudflare TURN fallback; k6 load tests sustained **200** concurrent WebSocket sessions and **27,000+** connections at **100%** handshake success
- Enforced campus-only access via @wisc.edu domain allowlisting at signup and JWT-validated WebSocket sessions, with per-user rate limiting to curb abuse

LLM-Based Database Schema Name Expansion – *Python, OpenAI API, spaCy*

2025

- Co-built an LLM pipeline that expands cryptic database column names (e.g., ModDate → Modified Date), achieving **76–88% accuracy** across 4 relational datasets
- Designed the evaluation harness that benchmarked all prompt variants: scoring LLM predictions against curated gold labels with precision/recall, synonym, and fuzzy matching